SOLEIL status update

Martin Savko

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Synchrotron SOLEIL

- energy: 2.75 GeV
- current: 500 mA
- electron beam lifetime: ~11 hours
- circumference: 354 m
- emittance (horizontal, vertical): 3.7×10^{-9}, 11×10^{-12} m.rad
- brilliance: 10^{20} ph.s^{-1}.$mrad^{-1}$.mm^{-2} @ 0.1% bandwidth

- founded in 2001, in operation since 2006
- funded jointly by CNRS (72%) and CEA (28%)
- 350 employees

Proxima 1

Source: **U20** in vacuum undulator

Focussing: KB, **CRL**, **20x40 µm, project for new KB mirrors**

Tunable: Si 111 CCM, 5.5 - 15.5 keV

Flux: **2.0e12 ph/s** @ 500mA @ 12.65keV

Area Detector: **Eiger X 16M**

XRF Detector: Ketek AXAS-M2 **H150** (XIA)

OAV Camera: Prosilica GC 1350 (4.65um, 1360x1024)

Goniometer: **SmarGon**

Sample Changer: CATS (**48 cryo, 16 ambient**) **Moving towards solution from SLS: Tell with a large dewar.**

MXCuBE: Qt4 v 2.3 (**CentOS 7**), HardwareRepository, Python 2.7. mxcubecore, mxcubeweb in development (nearly ready)

Proxima 2A

Source: **U24** in vacuum undulator

Focussing: KB, **horizontal PFM, 5x10 µm**

Tunable: Si 111 CCM, 5.5 - 18.5 keV

Flux: **1.6e12 ph/s** @ 500mA @ 12.65keV

Area Detector: **Eiger X 9M**

XRF Detector: Ketek AXAS-M2 **H80** (XIA, Xpress3)

OAV Camera: **BZoom**

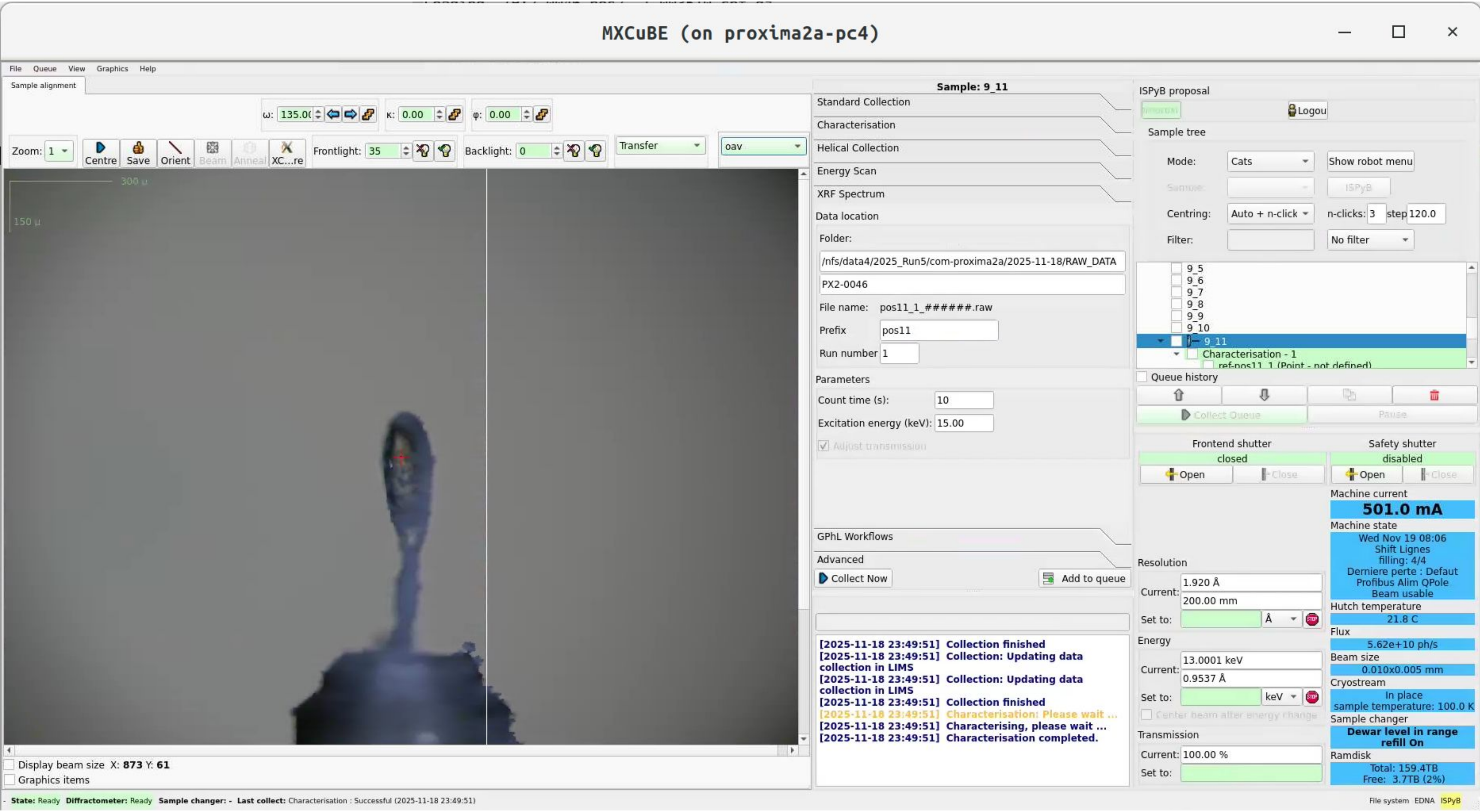
Goniometer: **MD3Up with minikappa (MK3 and HC/REX), Plate Screener**

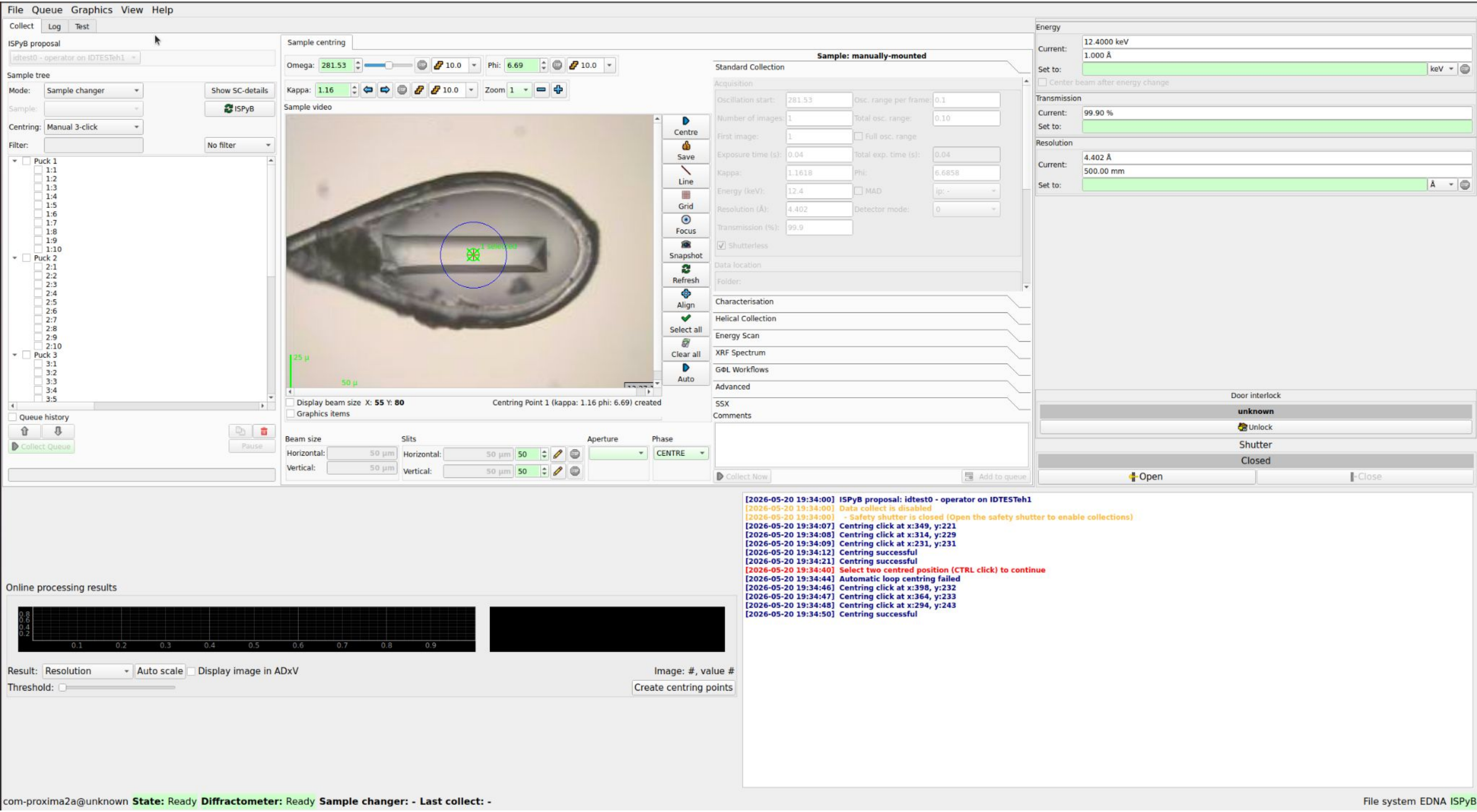
Sample Changer: CATS (**144 cryo, 48 ambient**)

MXCuBE: Qt5 (**Ubuntu 20.04**), mxcubecore, Python 3.8, Adating mxcubeqt to mxcubecore with AbstractDiffractometer, GPhL workflows, Will investigate Python >= 3.11

work done over the past 6 months

- unattended data collection
  - available on both beamlines (Murko, fast X-ray centring, volume reconstruction)
  - full ISPyB integration
  - speeding up the collects down to 2’15’’ per sample on Proxima2
    - mount: 40 s; opti: 25 s; tomo: 40s; char 60 s; main: ~30 s per sweep
  - groundwork for GPhL workflows integration into unattended mode
- mxcubeqt
  - PX2 mxcubecore with GenericDiffractometer, moving towards AbstractDiffractometer
  - PX1 based on HarwareRepository
- mxcubeweb
  - PX2 mxcubecore@AbstractDiffractometer mxcubeweb@
  - PX1 mxcubecore@GenericDiffractometer mxcubeweb@
    - Used internally, 2FA authentication implemented


MXCuBE (on proxima2a-pc4)
File Queue View Graphics Help
Sample alignment
ω: 135.0(κ: 0.00 φ: 0.00
Zoom: 1 Centre Save Orient Beam Anneal XC...re Frontlight: 35 Backlight: 0 Transfer oav
300 μ
150 μ
Display beam size X: 873 Y: 61
Graphics items
Sample: 9_11
Standard Collection
Characterisation
Helical Collection
Energy Scan
XRF Spectrum
Data location
Folder:
/nfs/data4/2025_Run5/com-proxima2a/2025-11-18/RAW_DATA
PX2-0046
File name: pos11_1_######.raw
Prefix pos11
Run number 1
Parameters
Count time (s): 10
Excitation energy (keV): 15.00
Adjust transmission
GPhL Workflows
Advanced
Collect Now
Add to queue
[2025-11-18 23:49:51] Collection finished
[2025-11-18 23:49:51] Collection: Updating data collection in LIMS
[2025-11-18 23:49:51] Collection: Updating data collection in LIMS
[2025-11-18 23:49:51] Collection finished
[2025-11-18 23:49:51] Characterisation: Please wait ...
[2025-11-18 23:49:51] Characterising, please wait ...
[2025-11-18 23:49:51] Characterisation completed.
ISPyB proposal
Logou
Sample tree
Mode: Cats
Show robot menu
ISPyB
Centring: Auto + n-click
n-clicks: 3 step 120.0
Filter: No filter
9_5
9_6
9_7
9_8
9_9
9_10
9_11
Characterisation - 1
ref-pos11_1 (Point - not defined)
Queue history
Collect Queue
Pause
Frontend shutter
closed
Open Close
Safety shutter
disabled
Open Close
Machine current
501.0 mA
Machine state
Wed Nov 19 08:06
Shift Lignes
filling: 4/4
Derniere perte : Defaut Profibus Alim QPole
Beam usable
Hutch temperature
21.8 C
Flux
5.62e+10 ph/s
Beam size
0.010x0.005 mm
Cryostream
In place
sample temperature: 100.0 K
Sample changer
Dewar level in range
refill On
Ramdisk
Total: 159.4TB
Free: 3.7TB (2%)
Resolution
Current: 1.920 Å
200.00 mm
Set to: Å
Energy
Current: 13.0001 keV
0.9537 Å
Set to: keV
Center beam after energy change
Transmission
Current: 100.00 %
Set to:
State: Ready Diffractometer: Ready Sample changer: - Last collect: Characterisation : Successful (2025-11-18 23:49:51)
File system EDNA ISPyB



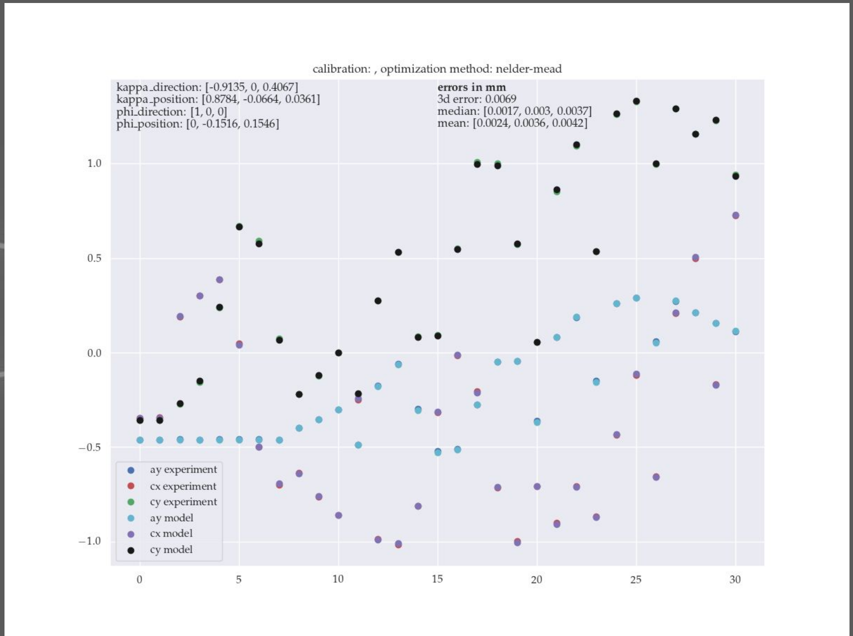
File Queue Graphics View Help
Collect
Log
Test
ISPyB proposal
Sample tree
Mode:
Sample changer
Show SC-details
ISPyB
Sample:
Centring:
Manual 3-click
Filter:
No filter
Puck 1
1:1
1:2
1:3
1:4
1:5
1:6
1:7
1:8
1:9
1:10
Puck 2
2:1
2:2
2:3
2:4
2:5
2:6
2:7
2:8
2:9
2:10
Puck 3
3:1
3:2
3:3
3:4
3:5
Queue history
Collect Queue
Pause
Sample centring
Omega:
281.53
10.0
Phi:
6.69
10.0
Kappa:
1.16
10.0
Zoom
1
Sample video
25 µ
50 µ
Centre
Save
Line
Grid
Focus
Snapshot
Refresh
Align
Select all
Clear all
Auto
Display beam size X: 55 Y: 80
Centring Point 1 (kappa: 1.16 phi: 6.69) created
Graphics items
Beam size
Horizontal:
50 µm
Vertical:
50 µm
Slits
Horizontal:
50 µm
50
Vertical:
50 µm
50
Aperture
Phase
CENTRE
Sample: manually-mounted
Standard Collection
Acquisition
Oscillation start:
281.53
Osc. range per frame:
0.1
Number of images:
1
Total osc. range:
0.10
First image:
1
Full osc. range
Exposure time (s):
0.04
Total exp. time (s):
0.04
Kappa:
1.1618
Phi:
6.6858
Energy (keV):
12.4
MAD
Resolution (Å):
4.402
Detector mode:
0
Transmission (%):
99.9
Shutterless
Data location
Folder:
Characterisation
Helical Collection
Energy Scan
XRF Spectrum
GΦL Workflows
Advanced
SSX
Comments
Collect Now
Add to queue
Energy
Current:
12.4000 keV
1.000 Å
Set to:
keV
Center beam after energy change
Transmission
Current:
99.90 %
Set to:
Resolution
Current:
4.402 Å
500.00 mm
Set to:
Å
Door interlock
unknown
Unlock
Shutter
Closed
Open
Close
[2026-05-20 19:34:00] ISPyB proposal: idtest0 - operator on IDTESTeh1
[2026-05-20 19:34:00] Data collect is disabled
[2026-05-20 19:34:00] - Safety shutter is closed (Open the safety shutter to enable collections)
[2026-05-20 19:34:07] Centring click at x:349, y:221
[2026-05-20 19:34:08] Centring click at x:314, y:229
[2026-05-20 19:34:09] Centring click at x:231, y:231
[2026-05-20 19:34:12] Centring successful
[2026-05-20 19:34:21] Centring successful
[2026-05-20 19:34:40] Select two centred position (CTRL click) to continue
[2026-05-20 19:34:44] Automatic loop centring failed
[2026-05-20 19:34:46] Centring click at x:398, y:232
[2026-05-20 19:34:47] Centring click at x:364, y:233
[2026-05-20 19:34:48] Centring click at x:294, y:243
[2026-05-20 19:34:50] Centring successful
Online processing results
0.8
0.6
0.4
0.2
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Result:
Resolution
Auto scale
Display image in ADxV
Image: #, value #
Threshold:
Create centring points
com-proxima2a@unknown State: Ready Diffractometer: Ready Sample changer: - Last collect: -
File system EDNA ISPyB

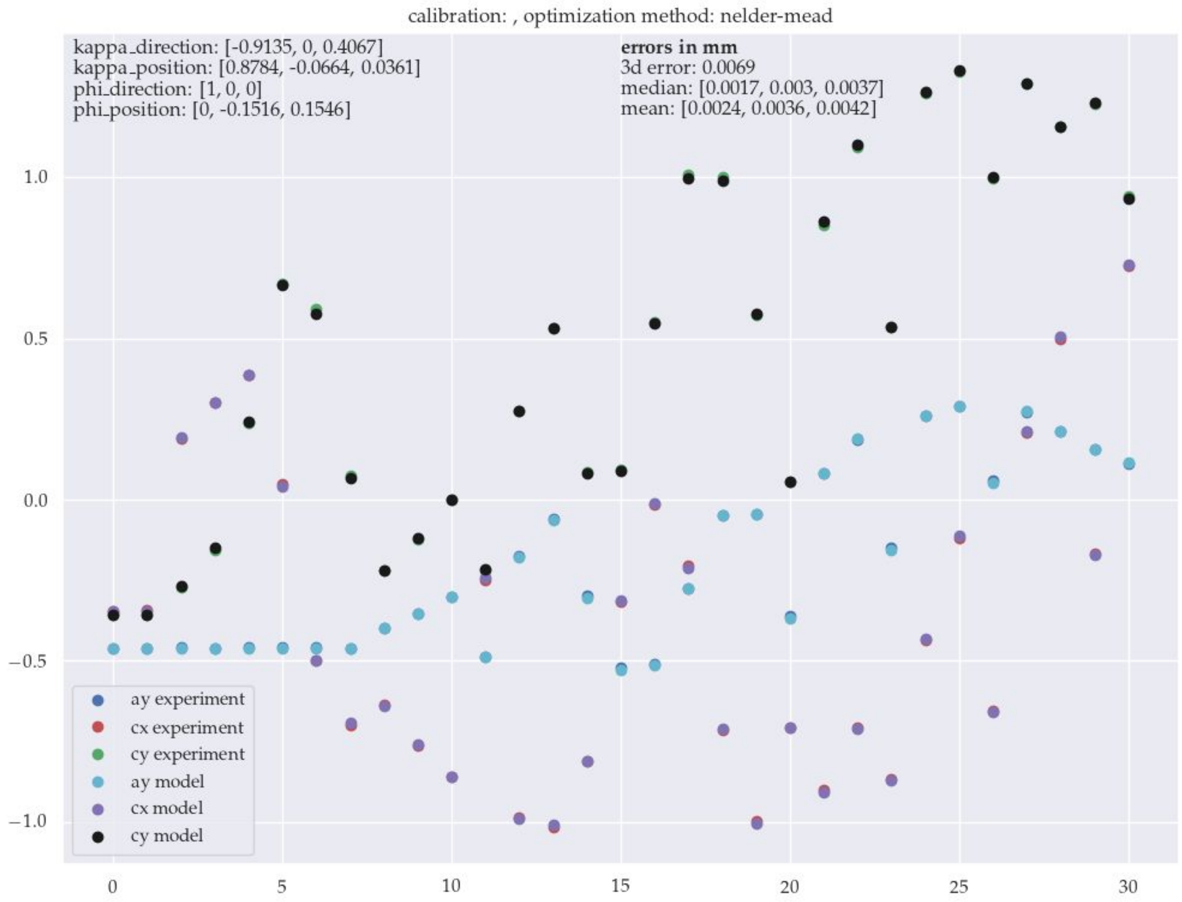
MD3 integration

- Excellent performance of the new goniometer
  - FATs Dec 2024; SATs Jan 2025
  - SOC ~ 150nm
  - MK3 Translation calibration < 4 um per axis
  - MK3 no longer necessary for sample exchange
    - was 180 deg, now 30 deg operating at this angle by default
  - Cryostream repositioned
  - Nearly instantaneous zoom changes.
  - Slipring issue December 2025

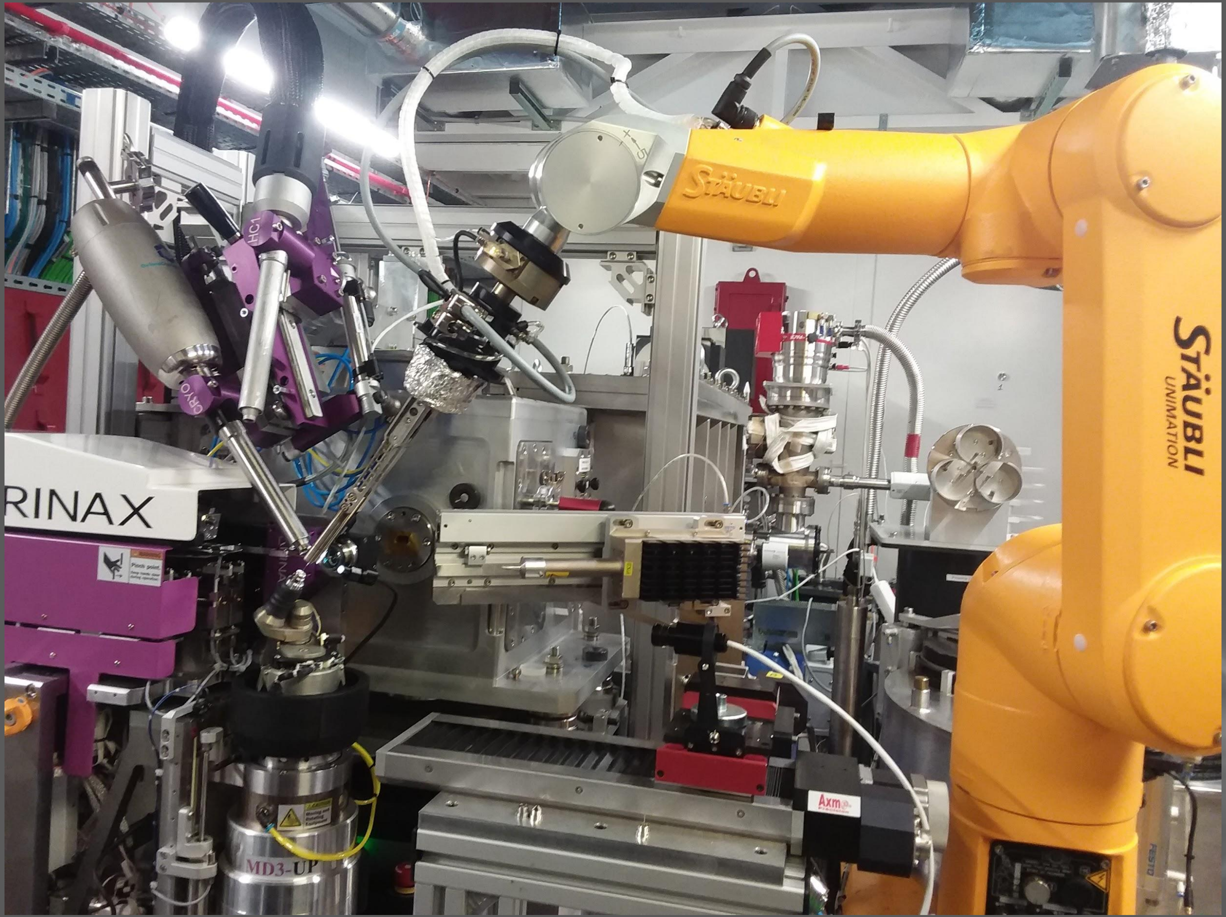
MK3 translation calibration

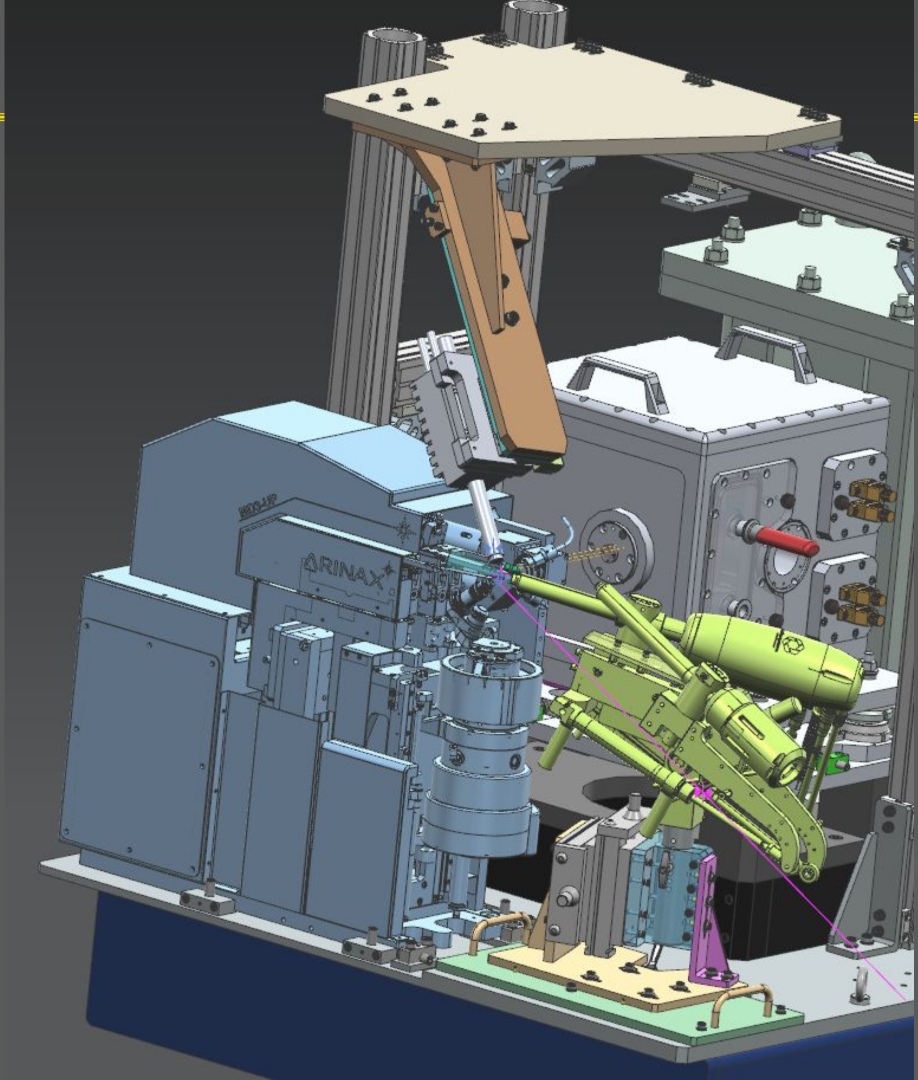
- Expected errors:
    - median errors: AlignmentY 1.7 um, CentringX 3.0 um, CentringY 3.7 um
    - 3d error: 6.9 um



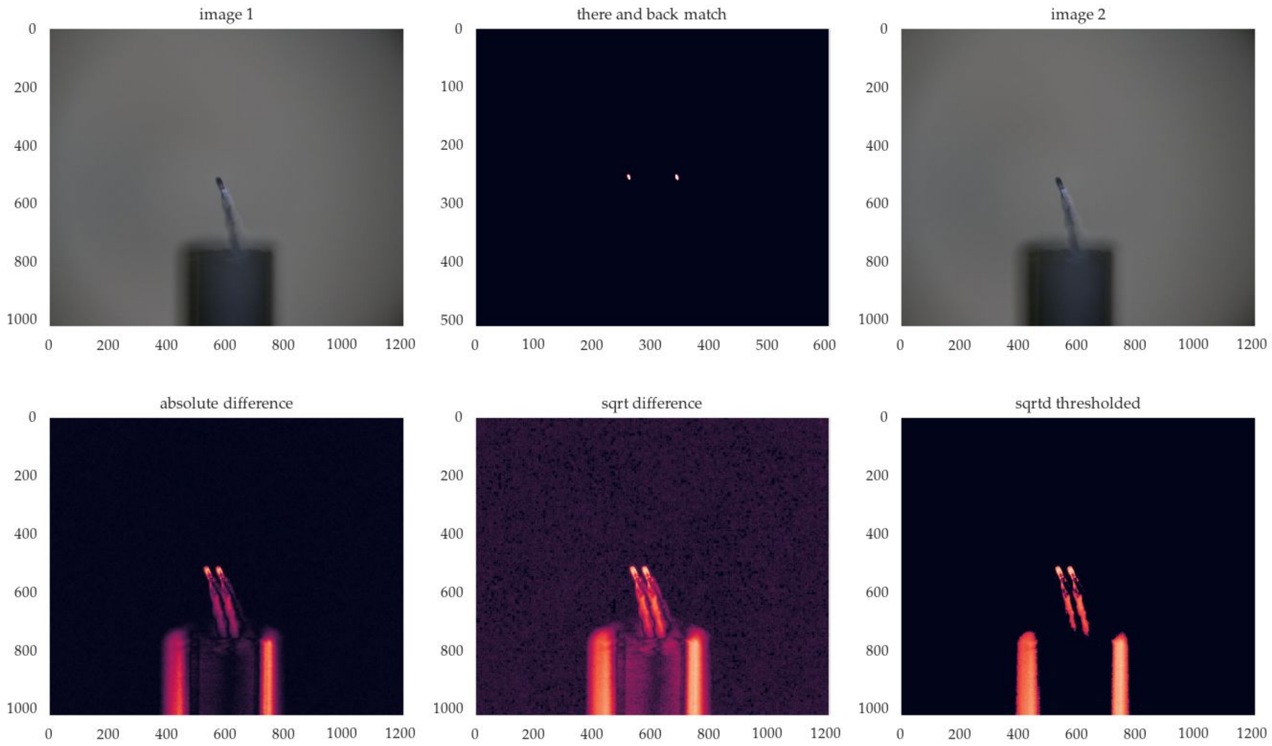

calibration: , optimization method: nelder-mead
kappa_direction: [-0.9135, 0, 0.4067]
kappa_position: [0.8784, -0.0664, 0.0361]
phi_direction: [1, 0, 0]
phi_position: [0, -0.1516, 0.1546]
errors in mm
3d error: 0.0069
median: [0.0017, 0.003, 0.0037]
mean: [0.0024, 0.0036, 0.0042]
1.0
0.5
0.0
−0.5
−1.0
0
5
10
15
20
25
30
ay experiment
cx experiment
cy experiment
ay model
cx model
cy model

Commissioning new goniometer!

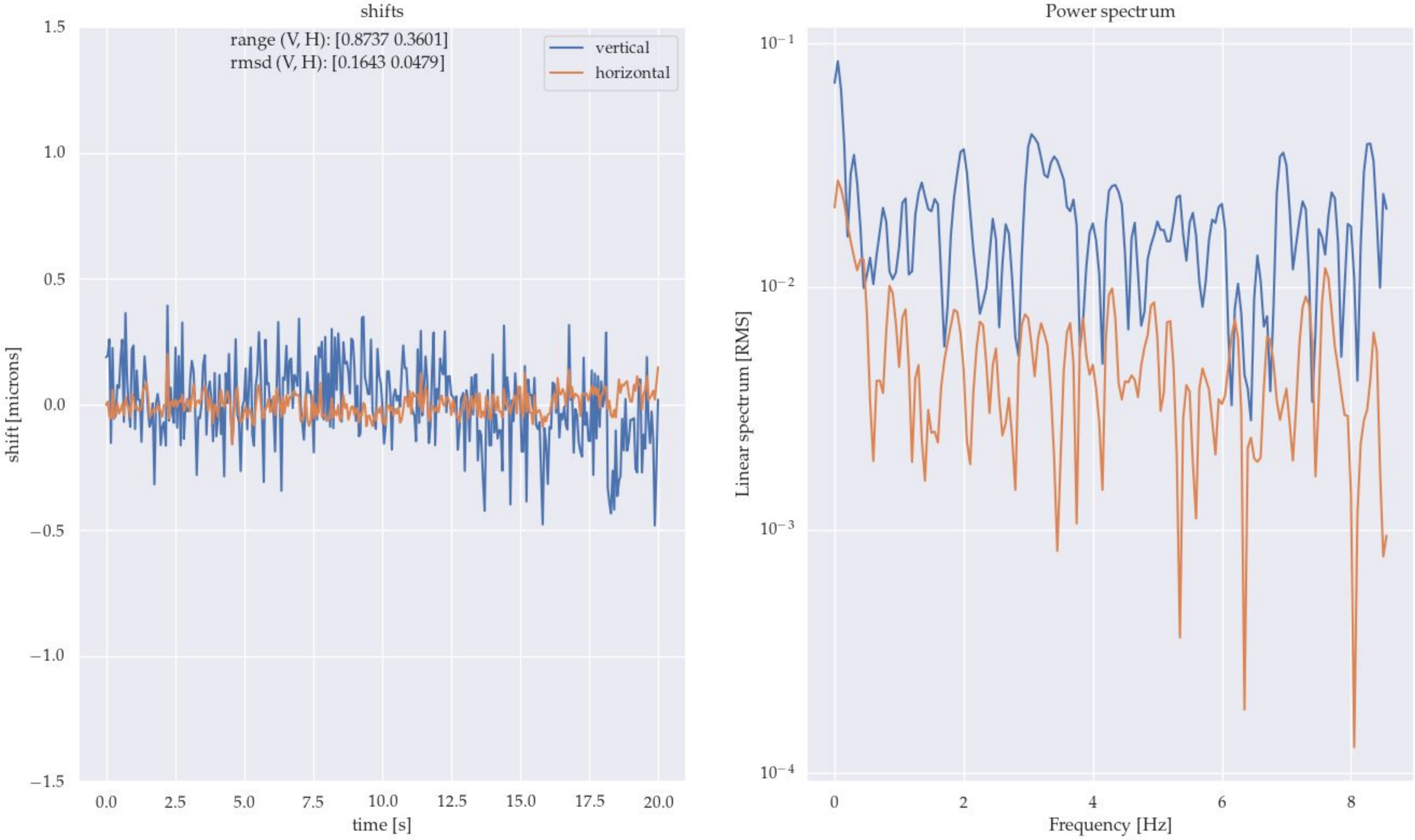



ARINAX

Sample stability

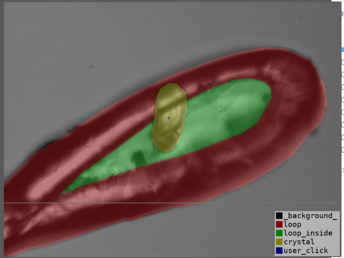


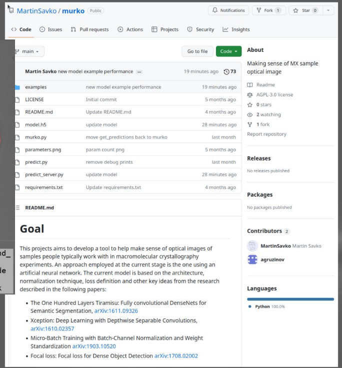
8_6_3, sampling 17.1 Hz



Plans for the next 6 months

- Finalizing mxcubeweb adaptation
  - Dan Costin
- Minikappa & GPhL workflows with MD3
  - mxcubeqt with AbstractDiffractometer
- Murko
  - update model, new tasks, larger dataset
- Volume aware experiments
  - rational spread of the dose (multi sweep experiments)





MartinSavko / murko Public

Code | Issues | Pull requests | Actions | Projects | Security | Insights

main · Go to file · Code

Martin Savko new model example performance · 19 minutes ago · 73

examples	new model example performance	19 minutes ago
LICENSE	Initial commit	5 months ago
README.md	Update README.md	4 months ago
model.h5	update model	28 minutes ago
murko.py	move get_predictions back to murko	last month
parameters.png	param count png	5 months ago
predict.py	remove debug prints	last month
predict_server.py	update model	28 minutes ago
requirements.txt	Update requirements.txt	4 months ago

README.md

Goal

This projects aims to develop a tool to help make sense of optical images of samples people typically work with in macromolecular crystallography experiments. An approach employed at the current stage is the one using an artificial neural network. The current model is based on the architecture, normalization technique, loss definition and other key ideas from the research described in the following papers:

- The One Hundred Layers Tiramisu: Fully convolutional DenseNets for Semantic Segmentation, arXiv:1611.09326
- Xception: Deep Learning with Depthwise Separable Convolutions, arXiv:1610.02357
- Micro-Batch Training with Batch-Channel Normalization and Weight Standardization arXiv:1903.10520
- Focal loss: Focal loss for Dense Object Detection arXiv:1708.02002

About

Making sense of MX sample optical image

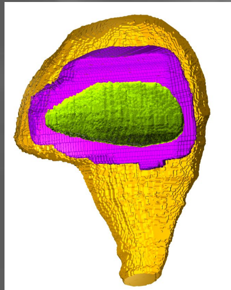
Readme · AGPL-3.0 license · 0 stars · 2 watching · 1 fork · Report repository

Releases: No releases published

Packages: No packages published

Contributors 2: MartinSavko Martin Savko · agruzinov

Languages: Python 100.0%



Acknowledgements

- SOLEIL team: Bill Shepard, Serena Sirigu, Rémi Soen, Damien Jeangerard, Eric Larquet, Pierre Legrand, Tatiana Isabet, Robin Lener, Andrew Thompson, Dan Costin, Arthur Felisaz-Calvino
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- SLS CF May Sharpe and Dennis Stagman
- EMBL HH team: Marina Novikova and Gleb Bourenkov
- Murko collaboration: Andrey Gruzinov and Thomas White (DESY), Didier Nurizzo (ESRF), Arthur Felisaz-Calvino (SOLEIL); David Aragao and Ralf Fleig (DLS), Annie Heroux & Nicola Demitru (Elettra), Jie Nan and Isak Lindé (Max IV), Roeland Boer (Alba), Tom Crosskey (Bessy), Kate Smith and Ezequiel Panepucci (formerly SLS), Scott Classen (ALS)